Subham Agrawal

B.Tech, Civil and Infrastructure Engineering
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SUMMARY

Final-year IIT Jodhpur undergraduate (minor in Mathematical and Computational Economics) who builds and backtests options and volatility models on real market data (NSE and SPX options, Binance order books), net of trading costs and out of sample. Seeking the Trainee – International Markets role.

EDUCATION

Degree	Institute	Year
B.Tech, Civil and Infrastructure Engineering	Indian Institute of Technology Jodhpur	2023 – 2027
Minor, Mathematical and Computational Economics	Indian Institute of Technology Jodhpur	2026 – 2027

Coursework: Financial Derivatives and Risk Management; Probability, Statistics and Stochastic Processes; Econometrics Lab; Games and Economic Behavior (Grade A); DSA; *ongoing:* ML for Economics, Intro to Machine Learning

PROJECTS

Nifty Options and India VIX: Fear Regimes and a Crash Hedge on Real NSE Prices 2026
FDRM course project, built solo, then extended to real NSE prices | *Python, pandas, statsmodels* App | GitHub | Real-data app | GitHub

- 1,536 days of Nifty 50 and India VIX (2020–26): VIX exceeded later realised volatility on 78% of days (Newey-West $t = 2.8$); re-implemented NSE's VIX formula and split the market into low, normal and high fear regimes (volatility 9%, 13%, 30%).
- On 1.2 million real Nifty option prices (NSE F&O bhavcopies), Black-Scholes (VIX + skew) overpriced 3-month 5% OTM puts by a median 33%.
- A VIX-timed protective put (buy below VIX 13, sell into spikes) returned 14.8% a year vs 11.1% unhedged at real prices, worst fall −22.8% vs −38.4%; most of the edge is one put held through the March 2020 crash (from July 2020: 16.2% vs 15.4%).

Deep Hedging of Options under Stochastic Volatility 2026
Independent project | *Python, PyTorch* Live app | GitHub

- Trained a neural network to hedge a short call with trading costs on Heston paths calibrated to SPX: cut CVaR95 by 33% vs a Heston delta plus variance-swap hedge on 100,000 unseen paths, trading less than half as much; on 16 years of real NIFTY 50 prices, 13% lower tail loss than Black-Scholes delta at 20 bp costs.

Heston Stochastic Volatility Pricing Engine 2026
Independent project | *Python, NumPy, SciPy* Live app | GitHub

- Seven pricing methods (Fourier, Monte Carlo, PDE) cross-checked against each other and published benchmarks; COS: 8e-8 error in 0.15 ms, about 10,000× faster than Monte Carlo. Calibrated to 1,880 SPX quotes at 0.90 vol-pt RMSE; misses short-dated skew (1.8 vol pts at 16 days).

DeepLOB: Predicting Price Moves from Live Crypto Order Books 2026
Independent project | *Python, PyTorch, ONNX* Live app | GitHub

- Recorded Binance BTC/ETH order books every 100 ms; beat chance (macro F1 0.54 at 10 s vs 0.33) but earned at most 0.9 bp a trade before costs vs a 10 bp taker fee, so not tradable (preliminary, 31 hours of data).

ML Stock Selection on Nifty: Catching Look-Ahead Bias 2026
Independent project | *Python, LightGBM, scikit-learn* Live app | GitHub

- Showed shuffled cross-validation inflated Sharpe 7× (3.68 vs 0.53 walk-forward), proved the leak with a placebo test on random walks and fixed it with purged cross-validation.

Landslide Susceptibility Mapping *B.Tech project:* 4,431 events; random splits leaked (AUC 0.97 vs 0.85); SVM AUC 0.905 App | GitHub

COMPETITIONS AND TRAINING

Inter IIT Tech Meet 13.0 (Zelta Labs) *Quant team member:* crypto price-direction models; team ranked 11th overall 2024
VC Analyst Case Study (Fin Maverick) *team ranked 4th of 159; picked 3 of 20 startups for a USD 10M fund* 2026
Quantitative Finance Cohort'24 *Quant Delegate:* stochastic calculus, options pricing, volatility models (Certificate) 2024

ACHIEVEMENTS

- **JEE Main 2023:** top 1% of 1.3 million. **IOQM 2020:** top 300 in Odisha (Certificate). **GATE 2026:** qualified (Civil).
- **NDA 2021:** qualified the written exam (Certificate). **Matiks:** 1500+ rating in maths sprint duels. **CBSE X:** 92%.
- **WorldQuant Challenge 2024:** Bronze level (Certificate). **IICPC 2024:** Competitive Programming Camp (Cert.).

TECHNICAL SKILLS

- **Derivatives:** futures and options, Black-Scholes, Greeks, put-call parity, implied vs realised volatility, delta hedging
- **Probability and Statistics:** conditional probability, expectation, hypothesis tests, regression, time series
- **Programming:** Python (pandas, NumPy, SciPy, statsmodels, scikit-learn, PyTorch, LightGBM), SQL, C++, Excel

POSITIONS OF RESPONSIBILITY

Assistant Head of Finance, Prometeo'25 *budgeted events and artists: about INR 15 lakh of an INR 40 lakh+ budget* 2025
Head of Media, Prometeo'26 *led 25+ juniors; several reels crossed 500k views* 2025 – 2026
Cultural Captain, Inter IIT Cultural Meet 7.0 and 8.0 *led a 25+ member contingent; top-10 finishes* 2024 – 2025
Coordinator, Frame-X Filmmaking Society *mentored 50+ students through workshops* 2024 – 2025